

Maryam Fathi M.Sc.

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Highly motivated and detail-oriented Ph.D. candidate in Pharmacology and Experimental Therapeutics at Boston University, with a strong molecular and cellular biology background. Passionate about drug discovery, molecular tool development, and cutting-edge genetic technologies. Experienced in CRISPR-based genome editing and next-generation sequencing (NGS). Adept at designing and executing experiments to investigate molecular mechanisms. Enthusiastic about advancing innovative biomedical solutions through rigorous research and technological advancements.

Current position

Graduate Candidate, Pharmacology and Experimental Therapeutic
Ignaty Leshchiner lab, Boston University, Boston, MA, USA

Sep 2023-Present

My research focuses on developing and applying advanced genomic, computational, and AI approaches to study tumor evolution using patient samples. I am developing a long-read scDNA and methylation sequencing method to detect rare somatic mutations, resolve cell-specific epigenetic states, and reconstruct clonal relationships within tumors. By integrating genomic, epigenomic, and other multi-omics data, I investigate tumor heterogeneity, subclonal dynamics, and the molecular events associated with cancer progression and metastasis, with the goal of identifying clinically relevant biomarkers and therapeutic vulnerabilities.

Education

M.Sc. Cell and Molecular Biology
Tehran Azad University, Tehran, Iran

2013-2015

Master dissertation project: Development of a high-throughput method compatible with a 'form invariant' protease activity assay for rapidly identifying metastasis biomarkers.

Project summary

Matrix Metalloproteinases (MMPs) have potential effects on tissue repair, cell invasion, and metastasis. The clinical detection of MMPs is used as biomarkers for various disorders, including cancer. Typically, proteases are identified using conventional antibody-based methods (such as ELISA or IHC) or matrix degradation-based assays. However, these assays fail to provide a readout of the proteases' bioactivity. Developing 'form invariant' detection platforms can potentially overcome this limitation. Unlike zymography and antibody-based techniques, this assay can measure the total activity of proteins over different time intervals.

B.Sc. Cell and Molecular Biology
Azad University, Arsanjan, Iran

2007-2011

Research Experiences

NSF Research Internship Award
Novartis Institutes for Biomedical Research, Cambridge, MA, USA

July to September 2025

I deployed advanced technologies and models to deliver components intracellularly, contributing to the advancement of therapeutic discovery.

Research Internship

Pfizer, Cambridge, MA, USA

July to September 2024

I employed functional genomic techniques to assess several targets in primary human B and T cells. This approach aimed to explore key pathways involved in B and T cells activation and proliferation during inflammation.

Research Associate

McKenna lab, Dartmouth College, Lebanon, NH, USA

June 2019-August 2023

Project summary

In the McKenna lab, I worked to enhance existing lineage tracing technologies using advanced CRISPR and base-editing methods. Most chemotherapeutic drugs are designed to stop the cell cycle progression of cancer cells; however, resistance to conventional chemotherapeutic agents continues to be a significant challenge in clinical cancer care. To address these issues and ultimately save patients' lives, we aimed to refine current lineage tracing technology to better understand and track tumor initiation and progression and discover new mechanisms of drug resistance in tumor cells.

Research Assistant (Volunteer)

Raman lab, Dartmouth College, Lebanon, NH, USA

Jan-June 2019

Project summary

In the Raman lab, I worked on understanding how the epithelial-to-mesenchymal transition (EMT) contributes to tumor metastasis and its consequences for drug resistance. We focused on inhibiting EMT or inducing mesenchymal-to-epithelial transition (MET) to increase tumor sensitivity to chemotherapeutic drugs and reduce metastasis.

Work Experience

Research Assistant, Biochemistry Quality Analyst

Fortress Company, Shiraz, Iran.

2012-2014

(Fortress is a diagnostics manufacturer based in the United Kingdom, specializing in the development, manufacturing, and distribution of a range of Medical Diagnostic and Point of Care Tests for a global market).

Selected Training and Professional Development

Selected Participant, Big Data Training for Cancer Research (BigCARE).

Chao Family Comprehensive Cancer Center, University of California, Irvine, CA, USA

July 2026

Selected for an intensive NCI-funded program focused on computational analysis, visualization, and integration of large-scale cancer omics data.

PUBLICATION

Fathi M, Cook A, Bagheri M, Curiel T, McKenna A. PALINCODE: Recording cell lineage with ternary palindromic CRISPR bits. *bioRxiv*. 2026. doi:10.64898/2026.04.16.718941.

TECHNICAL SKILLS

Microbiology Techniques

- Common bacterial experiments
- Bacterial transformation
- Bacterial electroporation
- Maintaining bacterial strain stocks

Bioinformatics & Data Analysis

- R
- Python
- FlowJo
- Fiji

Model Organisms

- Human cells
- Mouse

Genomics

- Next-generation sequencing (NGS)
Oxford Nanopore, Illumina
- NGS Library preparation
- CRISPR/Cas9 technology
- Single cell RNA sequencing
- sgRNA, shRNA, and siRNA design

Molecular biology Techniques

- PCR
- Complex cloning (Gibson and Golden gate assembly)
- Gel electrophoresis
- Western blotting
- ELISA
- DNA, RNA and Plasmid isolation
- Protein purification
- Biochemical assays
- Immunofixation

Tissue culture

- Common TC culture work
 - Transfection
 - Transduction
 - Electroporation
 - Immune cell (T-Cells and B-Cells)
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